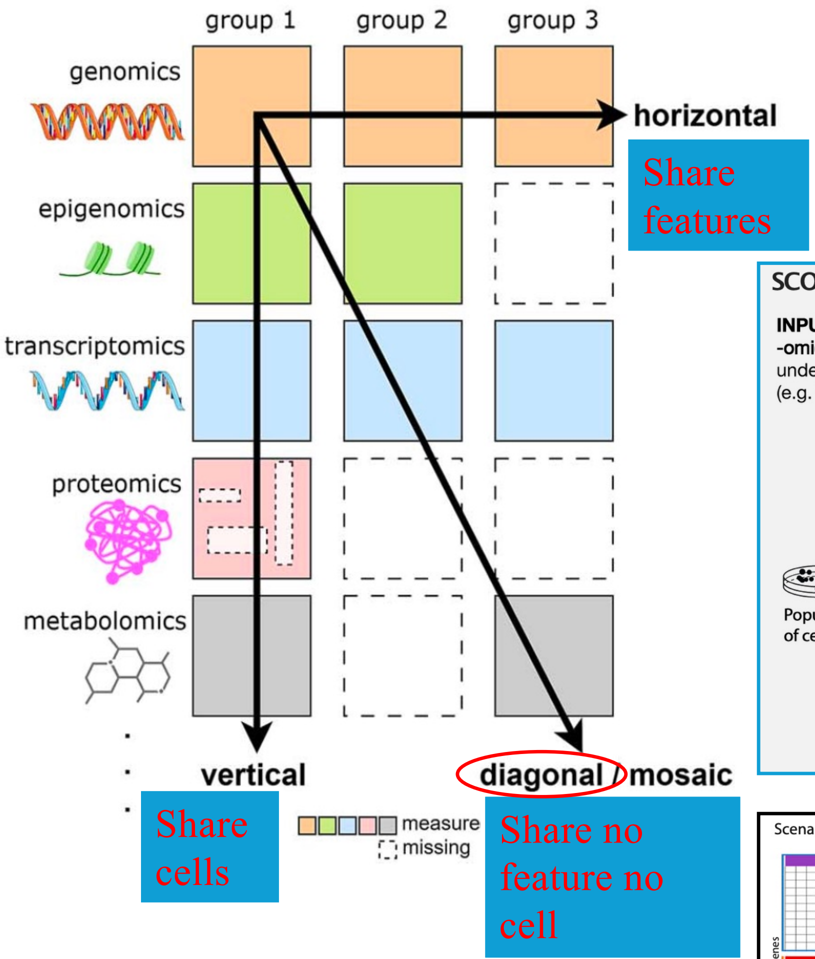
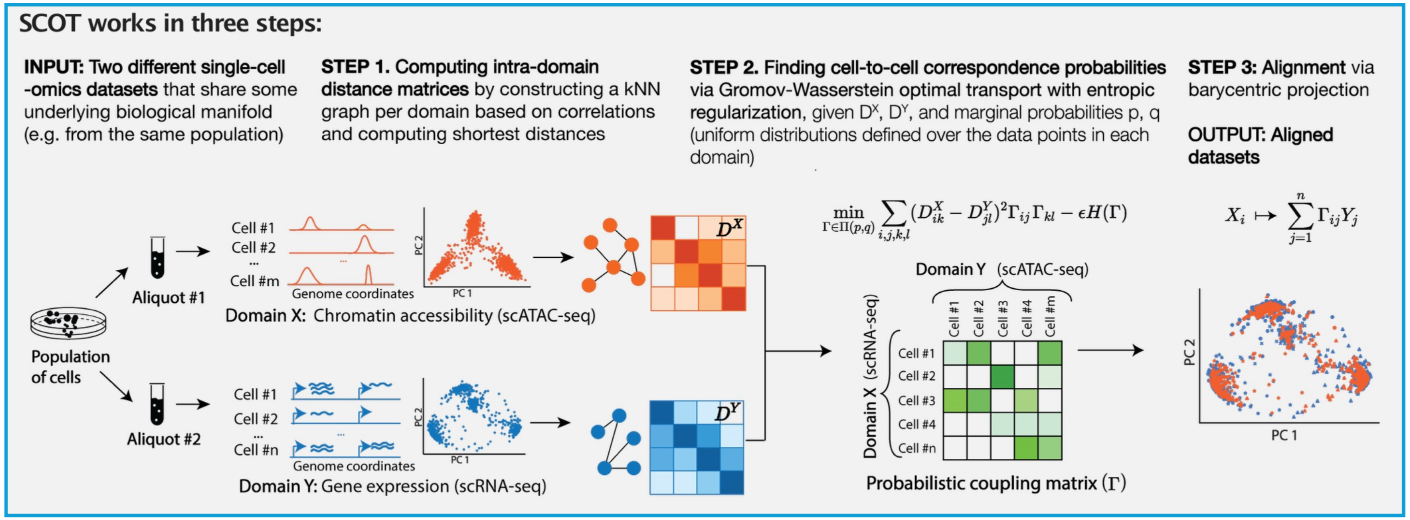
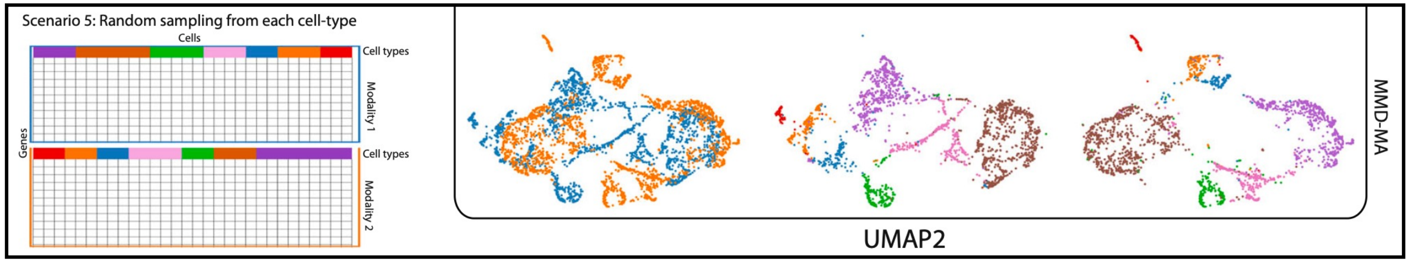
Pure diagonal integration:

* Share no feature no cell
* **Topological alignment**
* Assume similar/shared celltype structures
* May perform poorly
* Make more sense for spatial data (PASTE2, STalign)
* e.g. MMD-MA, PAMONA, SCOT, UniCom.



Baião et al. 2025 Briefings in Bioinformatics





Xu and McCord 2022 Nature communication

Prior knowledge guidance: Mosaic integration

- **Linked features** (partially share features – horizontal integration)
  - ❖RNA+ATAC (strong)
  - ❖RNA+surfaceProtein (weak)
  - ➢**Use only linked features**:
    - ➢E.g. Seurat_CCA/rPCA, LIGER-iNMF
  - ➢**Linked+unlinked features**:
    - ➢E.g. scMODEL, GLUE ,
    - ➢scVI, scANVI, MultiVI, LIGER-UINMF

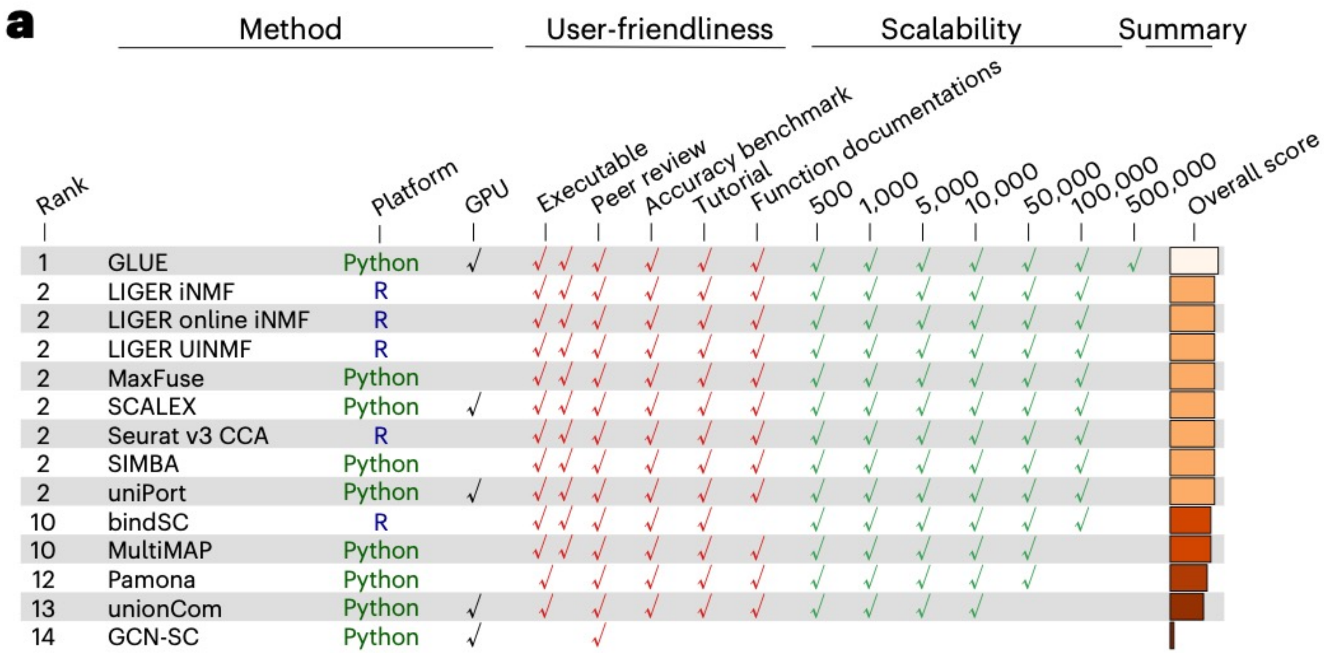
- **Leverage joint-profiling** (partially share cells – vertical integration)
  - ➢E.g. MIDAS, Seurat-bridge , Stabmap,
  - ➢MultiVI, Cobolt, Multigrate, scMoMAT, scVAEit

- **Leverage similar histology in spatial multi-omics**
  - ➢spatialEx+

Linear
NMF
Autoencoder

Linked features

scRNA+scATAC
- scATAC summarized to gene activity
- Regulatory interactions: feature correspondence



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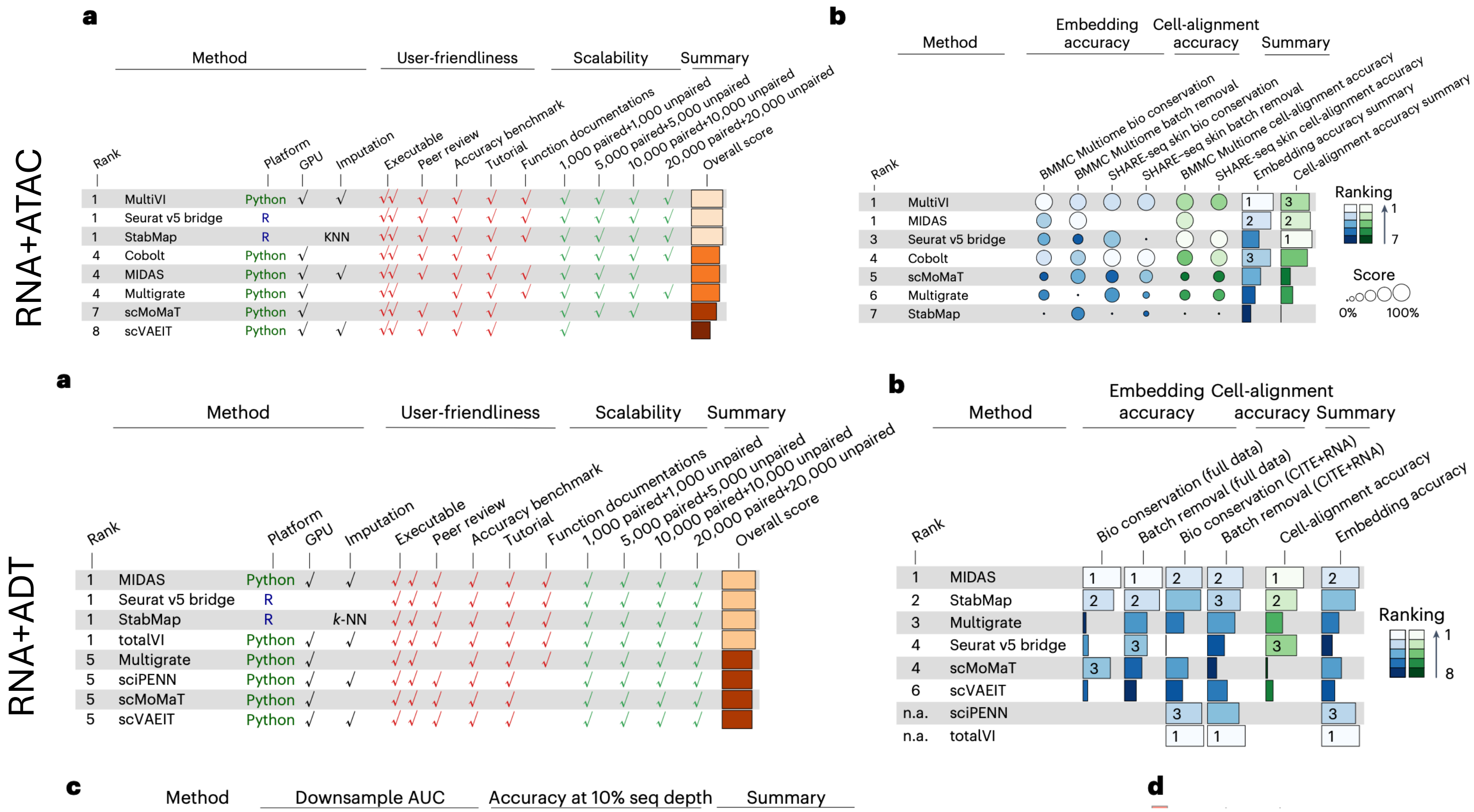
Rank	Method	Platform	GPU	Executable	Peer review	Accuracy benchmark	Tutorial	Function documentations	500	1,000	5,000	10,000	50,000	100,000	500,000	Overall score
1	GLUE	Python	√	√√	√	√	√	√	√	√	√	√	√	√	√	
2	LIGER iNMF	R		√√	√	√	√	√	√	√	√	√	√	√		
2	LIGER online iNMF	R		√√	√	√	√	√	√	√	√	√	√	√		
2	LIGER UINMF	R		√√	√	√	√	√	√	√	√	√	√	√		
2	MaxFuse	Python		√√	√	√	√	√	√	√	√	√	√	√		
2	SCALEX	Python	√	√√	√	√	√	√	√	√	√	√	√	√		
2	Seurat v3 CCA	R		√√	√	√	√	√	√	√	√	√	√	√		
2	SIMBA	Python		√√	√	√	√	√	√	√	√	√	√	√		
2	uniPort	Python	√	√√	√	√	√	√	√	√	√	√	√	√		
10	bindSC	R		√√	√	√	√		√	√	√	√	√	√		
10	MultiMAP	Python		√√	√	√	√	√	√	√	√	√	√			
12	Pamona	Python		√	√	√	√	√	√	√	√	√	√			
13	unionCom	Python	√	√	√	√	√	√	√	√	√	√				
14	GCN-SC	Python	√		√											

Method — User-friendliness — Scalability — Summary

Fu et al. 2025 Nature methods. https://doi.org/10.1038/s41592-025-02737-9

Leverage joint-profiling



Fu et al. 2025 Nature methods. https://doi.org/10.1038/s41592-025-02737-9